

CSDS 302 Discrete Math | Fall 2026 | Homework 3

Instructions

Section 4.1: 9, 11, 25

Section 4.2: 13, 27

Section 4.3: 2, 7

Section 4.4: 5, 16, 21, 35

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Feedback

Score: 86/88 (97.73%)

Section 4.1

9: -2, Just need to provide an example
for proof of existence

Section 4.1

9 Prove the statement: There is a real number x such that $x > 1$ and $2^x > x^{10}$.

Assume that $f(x) = 2^x - x^{10} > 0$. So we're looking for the value of x which $f(x) > 0$ at $x = 1$.

Since,

$$\begin{aligned} f(1) &= 2^1 - 1^{10} \\ f(1) &= 1 > 0 \end{aligned}$$

We can see that $f(x)$ is positive for $x = 1$.

Let's examine that derivative of $f(x)$ to understand the behaviour so, derivative of $f(x)$ is $f'(x)$,

$$f'(x) = 2^x \ln(2) - 10x^9$$

At $x = 1$:

$$f'(1) = 2 \ln(2)$$

$f(x)$ is increasing around $x = 1$. Since $f(x)$ is positive at $x = 1$ and $f(x)$ is increasing around $x = 1$ then there's a number x slightly greater than 1 where $f(x) > 0$ in essence, $2^x > x^{10}$

There exists a real number x such that $x > 1$ and $2^x > x^{10}$. This can be deduced by analyzing the function and its derivative. Since $f(1) > 0$ and $f'(x)$ is positive, then there must be a value slightly greater than 1 where $f(x) > 0$ fulfilling the above inequality.

11 There is an integer n such that $2n^2 - 5n + 2$ is prime.

Consider $n = 3$:

$$\begin{aligned} 2(3)^2 - 5(3) + 2 \\ = 18 - 15 + 2 \\ = 5 \end{aligned}$$

Since we found at least one integer ($n = 3$) for which the given expression evaluates to a prime number, we have proven the existence of such an n . Thus the statement is true.

25 Rewrite the statement with the quantification implicit as "If __, then __". Then, write the first sentence of a proof and the last sentence of a proof.

For all integers m and n , if $mn = 1$ then $m = n = 1$ or $m = n = -1$.

$\forall m, n \in \mathbb{Z}$, if $mn = 1$ then $m = n = 1$ or $m = n = -1$

If the product of two integers m and n is 1, then both m and n must be either 1 or -1.

Let m and n be the integers such that $mn = 1$. Therefore, the only integer solutions to $mn = 1$ are $m = 1, n = 1$, as required.

Section 4.2

13 Prove the statement "There exists an integer n such that $6n^2 + 27$ is prime" is false.

$$6n^2 + 27$$

$$3(2n^2 + 9)$$

The binomial has a common factor of 3. Thus, whatever the value of integer n , $6n^2 + 27$ will give us a value which is divisible by 3. And since a number which has more than 1 factor other than itself is a composite number, therefore there are no integer n that will make it a prime number. The statement is false.

27 Determine whether the statement is True or False. Justify your answer with a proof or a counterexample, as appropriate. In each case, use only the definitions of the terms and the Assumptions listed on page 161, not any previously established properties.

Statement: The difference of any two odd integers is even.

An odd integer is defined as $2k + 1$ for some integer k and an even integer is an integer that can be written in the form $2m$ for some integer m .

Let a and b be 2 arbitrary odd integers. By definition, we can write $a = 2k + 1$, $b = 2m + 1$, for some integers k and m . Now, we compute their difference:

$$\begin{aligned} a - b &= (2k + 1) - (2m + 1) \\ &= 2k - 2m \\ &= 2(k - m) \end{aligned}$$

Since $k - m$ is an integer, we can see that $a - b$ is of the form $2(\text{integer})$ which means $a - b$ is even.

We can conclude that the statement is true. For any two odd integers, their difference is always even, the statement is true.

Section 4.3

$$2 \frac{46,037}{10,000}$$

$$7 \frac{476927}{9090}$$

Section 4.4

5 Give a reason for your answer. Assume that all variables represent an integer. Is $6m(2m+10)$ divisible by 4?

$$6m(2m + 10)$$

$$12m(m + 5)$$

12 is divisible by 4. We check whether the remaining factors affect divisibility.

Let's say m is even and can be expressed as $m = 2k$ for some integer k , meaning:

$$12(2k)(m + 5) = 24k(m + 5)$$

Since 24 is divisible by 4, the whole expression is divisible by 4.

Let's say m is odd and can be expressed as $m = 2k + 1$ for some integer k , meaning:

$$12m(m + 5) = 12 \cdot \text{odd} \cdot \text{even}$$

Since 12 contains a factor of 4 and an even number is present, the product remains divisible by 4.

We can see that the expression $6m(2m + 10)$ is always divisible by 4 for any integer m , the statement is True.

16 For all integers a, b , and c , if $a|b$ then $a|c$ then $a|(b-c)$.

$a|b$, meaning there exists an integer k_1 such that $b = ak_1$.

$a|c$, meaning there exists an integer k_2 such that $c = ak_2$.

Prove $a|(b - c)$

Now compute $b - c$:

$$b - c = a(k_1 - k_2)$$

Since k_1 and k_2 are both integers, their difference $k_1 - k_2$ is also an integer. Since we can see that $b - c$ is a multiple of a , it follows that $a|(b - c)$.

21 Determine whether the statement is True or False. Prove the statement directly from the definitions if it is true, and give a counterexample if it is false.

Statement: The product of any two even integers is a multiple of 4.

Even integers can be expressed as $2k$ for some integer k . Assume we have two integers a and b , where $a = 2m$ and $b = 2n$ for some integers m and n .

$$\begin{aligned} a \cdot b &= (2m)(2n) \\ &= 4(mn) \end{aligned}$$

Since mn is an integer, we can rewrite this as:

$$a \cdot b = 4k, \text{ where } k = mn \text{ is an integer.}$$

This shows that the product of any two even integers is always divisible by 4.

The statement is true.

35 Two athletes run a circular track at a steady pace so that the first completes one round in 8 minutes and the second in 10 minutes. If they both start from the same spot at 4 p.m., when will be the first time they return to the start together?

First, we find the least common multiple of the two numbers 8 and 10. With prime factorization,

$8 = 2^3$ and $10 = 2 \cdot 5$. Taking the highest powers of all prime factors, we get, $2^3 \cdot 5 = 40$.

Thus, the first time both runners return to the start together is 40 minutes after 4:00pm, which is 4:40pm.